26. Bringing plant analysis to the next level

Supervisor(s): [redacted]

Description:

In a vertical farm, plants are grown in stacked layers within a closed environment, allowing for precise control of variables such as light and climate. This level of control enables systematic optimization of growing conditions, in which data science can play an important role.

To optimize the growth process and determine optimal environmental settings for a specific crop, reliable monitoring of plant growth is essential. Currently, plant growth is assessed through visual inspection or by destructive measurements taken after harvest. As these measurements destroy the plants, it is not possible to monitor growth continuously throughout the cultivation cycle, making it difficult to determine growth rates over time.

The goal of this project is to develop a non-destructive, automated method for monitoring plant growth using computer vision techniques. The project will explore multiple computer vision approaches in a vertical farm at the Brightlands Future Farming Institute (BFFI) on the Venlo campus.

Multiple 2D and 3D cameras will be used to collect images and depth maps of plants during their growth cycle. By combining depth information with semantic and instance segmentation of leaves, important plant traits such as height, biomass, and leaf size will be estimated. In parallel, climate parameters including temperature, humidity, air flow, and CO_2 concentration will be monitored and systematically varied between growth cycles. The resulting image-based measurements will be correlated with yield and growth performance.

Experiments will be conducted in controlled-environment chambers growing basil or lettuce.

In summary: you will be using sophisticated cameras and imaging techniques/state-of-the-art computer vision to build digital representations (twins) of plants, find optimal parameters for growing plants, and based on measured parameters find models that allow predicting the growth of the plants.

Requirements: Choose this project if you are deeply interested in state-of-the-art computer vision techniques. Good programming skills are required. We need team players as collaboration with our biologists and students from the University of Applied Sciences (HAS) will be key. You should be willing to occasionally travel to the Venlo campus to make optimal use of on-site facilities and expertise.

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